

Matrix Theory and Computation

Sections 2.1–2.3: Linear Algebra, Vector Norms, Matrix Norms

Detailed lecture notes for first-year postgraduate students

Golub & Van Loan, Chapter 2, pp. 63–75

Outline

1 §2.1 Basic Ideas from Linear Algebra

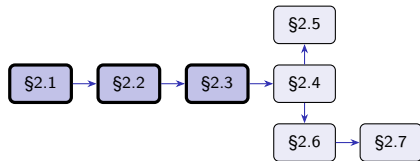
2 §2.2 Vector Norms

3 §2.3 Matrix Norms

Why This Chapter Matters

The purpose of Chapter 2

The analysis and derivation of algorithms in matrix computation requires a facility with linear algebra. **Norms are particularly important** — they are what let us say that a matrix is “nearly singular” or that an answer is “nearly right”.



The three sections covered here are the **foundation of the whole chain**:

- §2.1 supplies the vocabulary,
- §2.2 measures vectors,
- §2.3 measures matrices.

Everything after — the SVD, subspace metrics, sensitivity of $Ax = b$, and roundoff analysis — is built on them.

Learning Goals and Notation

By the end of these sections, you should be able to:

- 1 Compute a basis, a rank, and a null space by elimination.
- 2 Solve a low-rank updated system using previously computed solves.
- 3 Calculate vector and matrix norms and explain what they measure.
- 4 Derive a norm bound, check its assumptions, and compare it with an exact error.

$A \in \mathbb{R}^{m \times n}$ maps n input coordinates to m output coordinates. $A(:, j)$ is column j , e_j is a coordinate vector, and I is the identity.

Check: For a 3×2 matrix, what are the dimensions of x and Ax ?

A Suggested Three-Class Route

Class	Main tasks	Suggested time
1: linear algebra	Elimination, low-rank updates, orthogonality, eigenvalues	90 minutes
2: vector norms	Hand calculations, inequalities, error measures, convergence	60–75 minutes
3: matrix norms	Induced norms, spectral norm, Neumann series, perturbations	90 minutes

Slides labelled *Extension* can be used for tutorials or self-study. Work through the numbered calculation steps on the board before showing the answer.

Key point: The deck is a detailed teaching resource. Select examples to match the class pace.

Outline

1 §2.1 Basic Ideas from Linear Algebra

2 §2.2 Vector Norms

3 §2.3 Matrix Norms

Independence, Span, and Subspaces

Definition (Linear independence)

$\{a_1, \dots, a_n\}$ in $\mathbb{R}^m$ is *linearly independent* if $\sum_{j=1}^n \alpha_j a_j = 0$ implies $\alpha(1:n) = 0$. Otherwise it is *linearly dependent*.

Definition (Span)

$$\text{span}\{a_1, \dots, a_n\} = \left\{ \sum_{j=1}^n \beta_j a_j : \beta_j \in \mathbb{R} \right\}.$$

A *subspace* of $\mathbb{R}^m$ is a subset that is also a vector space; a span is always a subspace.

Key point: If $\{a_1, \dots, a_n\}$ is independent and $b \in \text{span}\{a_1, \dots, a_n\}$, then b is a *unique* linear combination of the a_j .

Example 1: A Redundant Set of Columns

Let

$$a_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}, \quad A = [a_1 \ a_2 \ a_3].$$

Step 1: find a dependence. Since $a_3 = a_1 + a_2$, $-a_1 - a_2 + a_3 = 0$ is a nontrivial relation.

Step 2: test the first two columns. If $\alpha a_1 + \beta a_2 = 0$, the third entry gives $\beta = 0$, and the first then gives $\alpha = 0$.

Step 3: describe the span.

$$\alpha a_1 + \beta a_2 = \begin{bmatrix} \alpha + 2\beta \\ 2\alpha + 4\beta \\ \beta \end{bmatrix} = \begin{bmatrix} s \\ 2s \\ t \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

Key point: Three listed vectors span a two-dimensional plane. The number of columns and the dimension are different quantities.

Subspace Tests and Direct Sums

A subspace

$$S = \{(s, 2s, t)^T : s, t \in \mathbb{R}\}.$$

Zero belongs to S . Sums and scalar multiples retain the relation $y_2 = 2y_1$. **An affine plane**

$$T = \{y : y_2 = 2y_1 + 1\}.$$

Zero does not belong to T , so T is not a subspace.

Check: For two subspaces, why does $S_1 \cap S_2 = \{0\}$ guarantee uniqueness?

A direct sum in $\mathbb{R}^2$

$$\mathbb{R}^2 = \text{span}\{e_1\} \oplus \text{span}\{e_2\}.$$

$(a, b)^T = ae_1 + be_2$ uniquely. For

$$S_1 = S_2 = \text{span}\{e_1\},$$

$$e_1 = 1e_1 + 0e_2 = 0e_1 + 1e_2.$$

The sum exists, but uniqueness fails.

Sums, Bases, and Dimension

Sums of subspaces

For subspaces $S_1, \dots, S_k$ of $\mathbb{R}^m$, the sum $S = \{a_1 + \dots + a_k : a_i \in S_i\}$ is a subspace, as is the intersection $S_1 \cap \dots \cap S_k$. The sum is a *direct sum*,

$$S = S_1 \oplus \dots \oplus S_k,$$

if each $v \in S$ has a **unique** such representation.

Basis and dimension

A *maximal linearly independent subset* $\{a_{i_1}, \dots, a_{i_k}\}$ of $\{a_1, \dots, a_n\}$ is one not properly contained in any independent subset. If maximal, $\text{span}\{a_1, \dots, a_n\} = \text{span}\{a_{i_1}, \dots, a_{i_k}\}$ and $\{a_{i_1}, \dots, a_{i_k}\}$ is a *basis*.

All bases for a subspace S have the same number of elements — the *dimension* $\dim(S)$.

Range and Null Space

Two subspaces are associated with an $m \times n$ matrix A .

Range (column space)

$$\text{ran}(A) = \{ y \in \mathbb{R}^m : y = Ax \text{ for some } x \in \mathbb{R}^n \}$$

Null space

$$\text{null}(A) = \{ x \in \mathbb{R}^n : Ax = 0 \}$$

If $A = [a_1 \mid \cdots \mid a_n]$ is a column partitioning, then

$$\text{ran}(A) = \text{span}\{a_1, \dots, a_n\}.$$

Key point: The range is spanned by the columns — this is why “range” and “column space” are the same thing.

Rank and the Rank–Nullity Relation

Definition (Rank)

$$\text{rank}(A) = \dim(\text{ran}(A)).$$

Equivalently, the maximal number of linearly independent columns (or rows).

Theorem (Rank–nullity)

If $A \in \mathbb{R}^{m \times n}$, then

$$\dim(\text{null}(A)) + \text{rank}(A) = n.$$

A is *rank deficient* if $\text{rank}(A) < \min\{m, n\}$.

Key point: Rank counts what survives the map; nullity counts what is destroyed. The two must add up to the dimension of the domain.

Example 1: Elimination and Rank

For the same matrix,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then $R_1 \leftarrow R_1 - 2R_2$ gives

$$R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There are two pivots: $\text{rank}(A) = 2$. A column-space basis uses *original* pivot columns:

$$\text{ran}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \right\}.$$

Key point: Row operations preserve linear relations among columns, but generally change the column space itself.

Example 1: Null Space and All Solutions

The equation $Rx = 0$ gives $x_1 + x_3 = 0$ and $x_2 + x_3 = 0$. Let $x_3 = t$. Then

$$x = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \text{null}(A) = \text{span} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Thus nullity = 1 and rank + nullity = 2 + 1 = 3.

For $b = (4, 8, 1)^T$, $x_p = (2, 1, 0)^T$ satisfies $Ax_p = b$. Every solution is

$$x = x_p + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - t \\ 1 - t \\ t \end{bmatrix}.$$

Check: Why can adding a null-space vector never change Ax ?

Example 1: A Rank Factorization

Since $a_3 = a_1 + a_2$, we can write

$$A = \underbrace{\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}}_{X \in \mathbb{R}^{3 \times 2}} \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}}_{Y^T \in \mathbb{R}^{2 \times 3}}.$$

Check the multiplication column by column.

$$X \begin{bmatrix} 1 \\ 0 \end{bmatrix} = a_1, \quad X \begin{bmatrix} 0 \\ 1 \end{bmatrix} = a_2, \quad X \begin{bmatrix} 1 \\ 1 \end{bmatrix} = a_3.$$

For any x , first form $c = Y^T x \in \mathbb{R}^2$, then form $Ax = Xc \in \mathbb{R}^3$.

Key point: A rank- r matrix can transmit its action through r intermediate coordinates. This is the algebra behind a linear low-rank layer.

The Matrix Inverse

If $A, X \in \mathbb{R}^{n \times n}$ satisfy $AX = I$, then $X = A^{-1}$ and A is *nonsingular*, otherwise A is *singular*.

Two identities used constantly

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (2.1.1)$$

$$(A^{-1})^T = (A^T)^{-1} \equiv A^{-T}. \quad (2.1.2)$$

Note the **reversal of order** in (2.1.1): the inverse of a product is the product of the inverses in the opposite order.

Key point: A^{-1} is a theoretical object. In computation we almost never form it — we factor A and solve.

Example 2: The Order in an Inverse Product

Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad AB = \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix}.$$

The individual inverses are

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}.$$

Correct order:

$$B^{-1}A^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ 0 & 1/3 \end{bmatrix}, \quad (AB)(B^{-1}A^{-1}) = I.$$

Reversing it gives $A^{-1}B^{-1} = \begin{bmatrix} 1/2 & -1/3 \\ 0 & 1/3 \end{bmatrix}$, which differs.

Check: Which cancellation is adjacent in $ABB^{-1}A^{-1}$?

The Sherman–Morrison–Woodbury Formula

How does the inverse change when the matrix changes? The starting identity is

$$B^{-1} = A^{-1} - B^{-1}(B - A)A^{-1}. \quad (2.1.3)$$

Woodbury formula [GVL (2.1.4)]

For $A \in \mathbb{R}^{n \times n}$ and U, V of size $n \times k$,

$$(A + UV^T)^{-1} = A^{-1} - A^{-1}U(I + V^T A^{-1}U)^{-1}V^T A^{-1},$$

assuming both A and $I + V^T A^{-1}U$ are nonsingular.

Key point: A correction of rank at most k changes the inverse by rank at most k , when both inverses exist.

The Sherman–Morrison Formula ($k = 1$)

The rank-one case is particularly useful. If $A \in \mathbb{R}^{n \times n}$ is nonsingular, $u, v \in \mathbb{R}^n$, and

$$\alpha = 1 + v^T A^{-1} u \neq 0,$$

then

$$(A + uv^T)^{-1} = A^{-1} - \frac{1}{\alpha} A^{-1} uv^T A^{-1}. \quad (2.1.5)$$

Why this is worth knowing

The right-hand side needs only **products with** A^{-1} , never a new factorization. If A has already been factored, a rank-one update costs $O(n^2)$ instead of the $O(n^3)$ of starting again.

Key point: The formula fails exactly when $\alpha = 0$ — precisely when $A + uv^T$ is singular.

Example 3: Deriving the Rank-One Solve

We want to solve $(A + uv^T)x = b$ using solves with A .

$$\begin{aligned}Ax + u(v^T x) &= b, \\x &= A^{-1}b - A^{-1}u(v^T x).\end{aligned}$$

Set $y = A^{-1}b$, $z = A^{-1}u$, and the scalar $s = v^T x$. Then

$$x = y - zs, \quad s = v^T y - (v^T z)s.$$

If $\alpha = 1 + v^T z \neq 0$,

$$s = \frac{v^T y}{\alpha}, \quad \boxed{x = y - z \frac{v^T y}{1 + v^T z}}.$$

Key point: Implementation: solve $Ay = b$, solve $Az = u$, form two dot products, then update one vector.

Example 3: Numerical Rank-One Update

Let $A = \text{diag}(2, 3)$, $u = v = (1, 1)^T$, and $b = (1, 2)^T$. Then $A + uv^T = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$.

$$Ay = b : \quad y = \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix},$$

$$\alpha = 1 + 1/2 + 1/3 = 11/6,$$

$$Az = u : \quad z = \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix},$$

$$v^T y = 1/2 + 2/3 = 7/6.$$

Therefore

$$x = \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix} - \frac{7}{11} \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} = \boxed{\begin{bmatrix} 2/11 \\ 5/11 \end{bmatrix}}.$$

Check: $\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 2/11 \\ 5/11 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$

Example 3: Inverse Update and a Small Denominator

For the preceding example,

$$(A + uu^T)^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} - \frac{6}{11} \begin{bmatrix} 1/4 & 1/6 \\ 1/6 & 1/9 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & -1 \\ -1 & 3 \end{bmatrix}.$$

A separate example shows why α matters. Take $A = I$, $u = e_1$, $v = -(1 - \varepsilon)e_1$. Then

$$\alpha = \varepsilon, \quad A + uv^T = \begin{bmatrix} \varepsilon & 0 \\ 0 & 1 \end{bmatrix}.$$

For $\varepsilon = 0$, the matrix is singular. For small nonzero ε , its inverse contains $1/\varepsilon$.

Key point: An algebraically valid update can still be sensitive. Check the update system and the final residual.

Example 4: Woodbury as a Small Linear System

For $(A + UV^T)x = b$, define

$$y = A^{-1}b, \quad W = A^{-1}U, \quad S = I_k + V^T W.$$

The computational steps are:

- 1 Solve $Ay = b$ and the k right-hand sides $AW = U$.
- 2 Form $S \in \mathbb{R}^{k \times k}$ and solve $Sc = V^T y$.
- 3 Return $x = y - Wc$.

Why it works. $x = y - W(V^T x)$, so $c = V^T x$ satisfies $Sc = V^T y$.

Key point: If $k \ll n$, reuse a factorization of A and solve a small k -by- k system. Do not explicitly invert S .

Example 4: A Complete Rank-Two Calculation

Take $A = 2I_3$, $U = V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, and $b = (1, 1, 0)^T$.

$$y = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

$$W = \frac{1}{2}U,$$

$$S = I_2 + \frac{1}{2}U^T U = \begin{bmatrix} 2 & 1/2 \\ 1/2 & 2 \end{bmatrix},$$

$$V^T y = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}.$$

The equations $2c_1 + c_2/2 = 1/2$ and $c_1/2 + 2c_2 = 1/2$ give $c_1 = c_2 = 1/5$.

$$x = y - Wc = \begin{bmatrix} 2/5 \\ 2/5 \\ -1/5 \end{bmatrix}, \quad \underbrace{\begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}}_{A+UU^T} x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Orthogonality

Definition

$\{x_1, \dots, x_p\}$ in $\mathbb{R}^m$ is *orthogonal* if $x_i^T x_j = 0$ whenever $i \neq j$, and *orthonormal* if $x_i^T x_j = \delta_{ij}$.

A set of nonzero, mutually orthogonal vectors is linearly independent. Zero vectors must be excluded from this independence claim.

Orthogonal complement

For a subspace $S \subseteq \mathbb{R}^m$,

$$S^\perp = \{y \in \mathbb{R}^m : y^T x = 0 \text{ for all } x \in S\}.$$

A relation used everywhere

$$\text{ran}(A)^\perp = \text{null}(A^T).$$

Orthogonal Matrices

$Q \in \mathbb{R}^{m \times m}$ is *orthogonal* if $Q^T Q = I$. If $Q = [q_1 \mid \cdots \mid q_m]$ is orthogonal, the q_i form an orthonormal basis for $\mathbb{R}^m$.

Theorem (2.1.1)

If $V_1 \in \mathbb{R}^{n \times r}$ has orthonormal columns, then there exists $V_2 \in \mathbb{R}^{n \times (n-r)}$ such that

$$V = [V_1 \mid V_2]$$

is orthogonal. Moreover $\text{ran}(V_1)^\perp = \text{ran}(V_2)$.

Key point: Any orthonormal set can always be *extended* to a full orthonormal basis. This is what makes “complete the basis” arguments legitimate — and it is a corollary of the QR factorization (§5.2).

Example 5: Constructing an Orthonormal Basis

Use $a_1 = (1, 2, 0)^T$, $a_2 = (2, 4, 1)^T$ from Example 1.

$$q_1 = a_1 / \|a_1\|_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix},$$

$$q_1^T a_2 = 10 / \sqrt{5} = 2\sqrt{5},$$

$$w_2 = a_2 - q_1(q_1^T a_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad q_2 = w_2.$$

Complete the basis with $q_3 = (-2, 1, 0)^T / \sqrt{5}$. Each q_i has unit length and $q_i^T q_j = 0$ for $i \neq j$.

$$Q = [q_1 \ q_2 \ q_3], \quad Q^T Q = I_3.$$

Key point: Normalization changes length; subtracting the projection removes the component along an existing basis vector.

Example 5: Projection onto a Line

Let $x = (3, 1)^T$ and $u = (1, 1)^T$. The closest point $p = tu$ minimizes

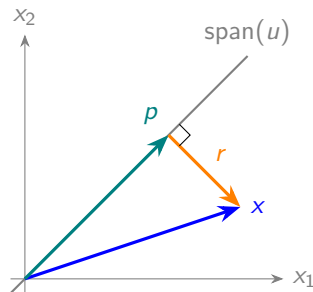
$$\|x - tu\|_2^2 = (3 - t)^2 + (1 - t)^2.$$

Differentiate: $4t - 8 = 0$, so $t = 2$.

$$p = (2, 2)^T, \quad r = x - p = (1, -1)^T, \quad u^T r = 0.$$

$$\text{Also } \|x\|_2^2 = 10 = 8 + 2 = \|p\|_2^2 + \|r\|_2^2.$$

Key point: $p = u(u^T x)/(u^T u)$. With a unit vector q , this becomes $p = qq^T x$.



Synthesis: The Four Fundamental Subspaces

For $A \in \mathbb{R}^{m \times n}$ of rank r :

Subspace	Lives in	Dimension	Orthogonal complement
$\text{ran}(A)$	$\mathbb{R}^m$	r	$\text{null}(A^T)$
$\text{null}(A^T)$	$\mathbb{R}^m$	$m - r$	$\text{ran}(A)$
$\text{ran}(A^T)$	$\mathbb{R}^n$	r	$\text{null}(A)$
$\text{null}(A)$	$\mathbb{R}^n$	$n - r$	$\text{ran}(A^T)$

Two orthogonal decompositions follow:

$$\mathbb{R}^m = \text{ran}(A) \oplus \text{null}(A^T), \quad \mathbb{R}^n = \text{ran}(A^T) \oplus \text{null}(A).$$

Key point: Rank-nullity plus $\text{ran}(A)^\perp = \text{null}(A^T)$ gives the entire picture. §2.4 will show that the SVD produces orthonormal bases for all four at once.

Example 1: The Four Subspaces Computed

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$, $\text{rank} = 2$:

Subspace	A basis
$\text{ran}(A)$	$(1, 2, 0)^T, (2, 4, 1)^T$
$\text{null}(A^T)$	$(-2, 1, 0)^T$
$\text{ran}(A^T)$	$(1, 2, 3)^T, (0, 1, 1)^T$
$\text{null}(A)$	$(-1, -1, 1)^T$

For example, $(1, 2, 3)(-1, -1, 1)^T = 0$ and $(0, 1, 1)(-1, -1, 1)^T = 0$.

Solvability test. $Ax = b$ is consistent exactly when $(-2, 1, 0)b = 0$. Thus $(4, 8, 1)^T$ is consistent, but $(4, 7, 1)^T$ is not.

Key point: The left null space supplies constraints that every attainable right-hand side must satisfy.

The Determinant, and Differentiation

Determinant [GVL §2.1.6]

$\det(A) = \sum_{j=1}^n (-1)^{j+1} a_{1j} \det(A_{1j})$, where A_{1j} deletes row 1 and column j . Then

$$\det(AB) = \det(A) \det(B), \quad \det(A^T) = \det(A), \quad \det(cA) = c^n \det(A),$$

and $\det(A) \neq 0$ **if and only if** A is **nonsingular**.

Differentiation [GVL §2.1.8]

If the entries of $A(\alpha)$ are differentiable, $\dot{A}(\alpha)$ is defined entrywise: $\dot{A}(\alpha) = (\dot{a}_{ij}(\alpha))$. Two standard consequences:

$$\frac{d}{d\alpha}(AB) = \dot{A}B + A\dot{B}, \quad \frac{d}{d\alpha}(A^{-1}) = -A^{-1}\dot{A}A^{-1}.$$

Key point: The determinant is useful theory but a poor numerical singularity test: $\det(cA) = c^n \det(A)$ scales arbitrarily without changing conditioning. Differentiation, by contrast, is a genuine *sensitivity* tool — the theme of §2.6.

Example 6: Determinant Size and Sensitivity

Small determinant, uniform scaling

$$A = 10^{-8}I_2, \quad \det(A) = 10^{-16}.$$

All vectors shrink by the same factor 10^{-8} .

$$\|A\|_2 \|A^{-1}\|_2 = 1.$$

The product $\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2$ is the condition number, introduced systematically in §2.6.

Key point: The determinant detects exact singularity. Its magnitude alone does not measure relative sensitivity.

Unit determinant, unequal scaling

$$B = \begin{bmatrix} 10^8 & 0 \\ 0 & 10^{-8} \end{bmatrix}, \quad \det(B) = 1.$$

One direction expands while another contracts.

$$\|B\|_2 \|B^{-1}\|_2 = 10^{16}.$$

Example 7: Differentiating an Inverse

Assume $A(t)$ is differentiable and nonsingular near the value of interest. Differentiate $A(t)A(t)^{-1} = I$:

$$A'(t)A(t)^{-1} + A(t)(A(t)^{-1})' = 0.$$

Multiplying on the left by $A(t)^{-1}$ gives

$$(A^{-1})' = -A^{-1}A'A^{-1}.$$

Numerical-symbolic check.

$$A(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}, \quad A(t)^{-1} = \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix}, \quad A' = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Both direct differentiation and $-A^{-1}A'A^{-1}$ give $\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$.

Extension: Differentiating a Quadratic Objective

Let $\phi(x) = \frac{1}{2}x^T Ax - b^T x$. In direction h ,

$$\phi(x + th) - \phi(x) = \frac{t}{2}(h^T Ax + x^T Ah) - tb^T h + O(t^2).$$

Since $x^T Ah = h^T A^T x$,

$$\nabla\phi(x) = \frac{1}{2}(A + A^T)x - b.$$

For $A = \begin{bmatrix} 2 & 3 \\ 0 & 4 \end{bmatrix}$, $b = (1, 2)^T$, and $x = (1, 1)^T$,

$$\nabla\phi(x) = \begin{bmatrix} 2 & 3/2 \\ 3/2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 7/2 \end{bmatrix}.$$

Key point: The familiar formula $Ax - b$ requires symmetry. This is the link to gradients in optimization.

Eigenvalues and Eigenvectors

The eigenvalues of $A \in \mathbb{C}^{n \times n}$ are the zeros of the *characteristic polynomial*

$$p(x) = \det(A - xI), \quad \lambda(A) = \{x : \det(A - xI) = 0\}.$$

Every $n \times n$ matrix has n complex eigenvalues, counting algebraic multiplicity.

If the eigenvalues are real we index them largest to smallest:

$$\lambda_n(A) \leq \cdots \leq \lambda_2(A) \leq \lambda_1(A),$$

writing $\lambda_{\max}(A) = \lambda_1(A)$ and $\lambda_{\min}(A) = \lambda_n(A)$.

If $\lambda \in \lambda(A)$ there is a nonzero x with $Ax = \lambda x$; such an x is an *eigenvector*.

Key point: Needed here mainly to support the SVD (§2.4), positive definiteness (§4.2), and fast solvers (§4.8) — the full treatment waits for Chapters 7–8.

Similarity and Diagonalizability

If $X \in \mathbb{C}^{n \times n}$ is nonsingular and $B = X^{-1}AX$, then A and B are *similar*.

Similar matrices have exactly the same eigenvalues.

If A has n independent eigenvectors $x_1, \dots, x_n$ with $Ax_i = \lambda_i x_i$, then A is *diagonalizable*: with $X = [x_1 \mid \dots \mid x_n]$,

$$X^{-1}AX = \text{diag}(\lambda_1, \dots, \lambda_n).$$

But not all matrices are diagonalizable

Diagonalizability requires a full set of independent eigenvectors — a condition that can fail.

Example 8: Repeated and Complex Eigenvalues

A repeated eigenvalue

$$J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \det(J - \lambda I) = (1 - \lambda)^2.$$

$(J - I)x = 0$ gives $x_2 = 0$. Only multiples of e_1 are eigenvectors, so no invertible eigenvector matrix exists.

Key point: Count eigenvalues with multiplicity. A full set of independent eigenvectors is an additional requirement.

A real matrix with complex eigenvalues

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \det(R - \lambda I) = \lambda^2 + 1.$$

Its eigenvalues are i and $-i$. It rotates real vectors by 90° , so no real line stays fixed.

The Symmetric Case

If $A \in \mathbb{R}^{n \times n}$ is **symmetric**, the difficulty disappears entirely: there exists an *orthogonal* Q with

$$Q^T A Q = \text{diag}(\lambda_1, \dots, \lambda_n). \quad (2.1.6)$$

Why symmetry is such good news

- The eigenvalues are **real**.
- The eigenvectors can be chosen **orthonormal**.
- The similarity transformation is **orthogonal**, so (as §2.3 will show) it does not change the 2-norm or the Frobenius norm — it cannot amplify error.

Key point: Symmetric matrices are always orthogonally diagonalizable. This single fact is why symmetric eigenproblems are far better behaved than general ones.

Example 9: Symmetric Diagonalization by Hand

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Then

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

For $\lambda = 3$, $x_1 = x_2$; for $\lambda = 1$, $x_1 = -x_2$. Normalize:

$$q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad Q = [q_1 \ q_2].$$

Check $Q^T Q = I$, $Aq_1 = 3q_1$, and $Aq_2 = q_2$. Therefore

$$Q^T A Q = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad A = Q \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} Q^T.$$

Check: Which direction is stretched most? What happens to the perpendicular direction?

The Rayleigh Quotient

For symmetric A , the extreme eigenvalues have a variational characterization:

$$\lambda_{\max}(A) = \max_{x \neq 0} \frac{x^T A x}{x^T x}, \quad (2.1.7)$$

$$\lambda_{\min}(A) = \min_{x \neq 0} \frac{x^T A x}{x^T x}. \quad (2.1.8)$$

The ratio $x^T A x / x^T x$ is the *Rayleigh quotient*.

Consequence

For every $x \neq 0$,

$$\lambda_{\min}(A) \|x\|_2^2 \leq x^T A x \leq \lambda_{\max}(A) \|x\|_2^2.$$

Key point: An eigenvalue problem is turned into an *optimization* problem. This is the bridge to §2.3, where $\|A\|_2$ is characterized the same way.

Example 9: Why the Rayleigh Bounds Hold

For any nonzero x , write $x = Qc$ in the orthonormal eigenbasis. Then

$$\frac{x^T A x}{x^T x} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2} = \sum_i \underbrace{\frac{c_i^2}{\sum_j c_j^2}}_{w_i \geq 0, \sum_i w_i = 1} \lambda_i.$$

It is a weighted average of the eigenvalues. Hence $\lambda_{\min} \leq R_A(x) \leq \lambda_{\max}$.

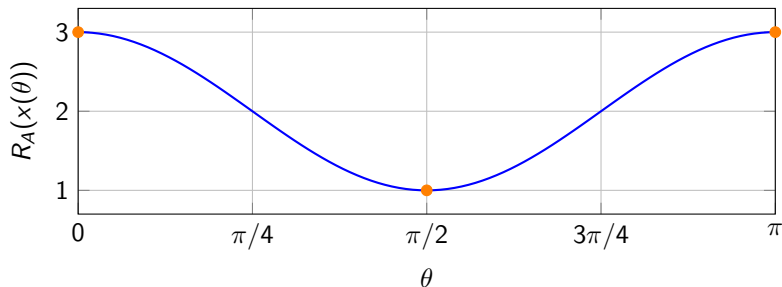
For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

$$R_A(e_1) = 2, \quad R_A((1, 1)^T) = 3, \quad R_A((1, -1)^T) = 1.$$

Key point: Extreme eigenvectors attain the bounds. A mixture of eigendirections gives an intermediate value.

Example 9: Rayleigh Quotient as a Curve

Set $x(\theta) = \cos \theta q_1 + \sin \theta q_2$. Then $\|x(\theta)\|_2 = 1$ and $R_A(x(\theta)) = 3 \cos^2 \theta + \sin^2 \theta$.



Key point: Changing the sign of an eigenvector does not change its Rayleigh quotient.

§2.1 — What to Carry Forward

Structure

- $\text{rank}(A) + \dim(\text{null}(A)) = n$
- $\text{ran}(A)^\perp = \text{null}(A^T)$
- Orthonormal sets extend to full bases (Thm 2.1.1)

Inverses

- $(AB)^{-1} = B^{-1}A^{-1}$
- Rank at most k update $\Rightarrow$ inverse change of rank at most k (2.1.4)

Spectra

- Similarity preserves eigenvalues
- Symmetric $\Rightarrow Q^T A Q$ diagonal, real spectrum
- Rayleigh quotient gives $\lambda_{\max}, \lambda_{\min}$

The through-line

Orthogonality is the recurring theme. It gives clean decompositions, well-behaved eigenproblems, and — next — norms that transformations cannot distort.

Outline

- 1 §2.1 Basic Ideas from Linear Algebra
- 2 §2.2 Vector Norms
- 3 §2.3 Matrix Norms

What a Norm Is For

A norm on a vector space plays the same role as absolute value: **it furnishes a distance measure.**

$\mathbb{R}^n$ together with a norm becomes a *metric space*, which is what makes the familiar notions — neighbourhood, open set, convergence, continuity — available.

Definition (Vector norm)

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a vector norm if

$$\begin{aligned} f(x) &\geq 0, & x &\in \mathbb{R}^n, \quad (f(x) = 0 \text{ iff } x = 0), \\ f(x+y) &\leq f(x) + f(y), & x, y &\in \mathbb{R}^n, \\ f(\alpha x) &= |\alpha|f(x), & \alpha &\in \mathbb{R}, \quad x \in \mathbb{R}^n. \end{aligned}$$

We write $f(x) = \|x\|$.

Key point: Positivity, the triangle inequality, homogeneity. Nothing else.

The p -Norms

A useful class of vector norms:

$$\|x\|_p = (|x_1|^p + \cdots + |x_n|^p)^{1/p}, \quad p \geq 1. \quad (2.2.1)$$

The three that matter

$$\|x\|_1 = |x_1| + \cdots + |x_n| \quad (\text{sum of magnitudes})$$

$$\|x\|_2 = (|x_1|^2 + \cdots + |x_n|^2)^{1/2} = (x^T x)^{1/2} \quad (\text{Euclidean length})$$

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i| \quad (\text{largest component})$$

A *unit vector* with respect to $\|\cdot\|$ satisfies $\|x\| = 1$.

Key point: For dimension at least two, the 2-norm is the only p -norm that comes from an inner product — which is exactly why it interacts so well with orthogonality.

Example 10: Computing and Normalizing a Vector

For $x = (3, -4, 0)^T$,

$$\|x\|_1 = |3| + |-4| + |0| = 7,$$

$$\|x\|_2 = \sqrt{3^2 + (-4)^2 + 0^2} = 5,$$

$$\|x\|_\infty = \max\{3, 4, 0\} = 4.$$

The word “unit” depends on the chosen norm:

$$\frac{x}{7} \text{ has 1-norm } 1, \quad \frac{x}{5} \text{ has 2-norm } 1, \quad \frac{x}{4} \text{ has infinity-norm } 1.$$

Check: If $q = x/5$, is $\|q\|_1 = 1$? Calculate $7/5$.

Example 11: Functions That Fail to Be Norms

Squared Euclidean length. Let $f(x) = \|x\|_2^2$. For $x = e_1$,

$$f(2x) = 4, \quad |2|f(x) = 2.$$

Homogeneity fails, although f is useful as a loss function.

The p expression with $p < 1$. For $p = 1/2$,

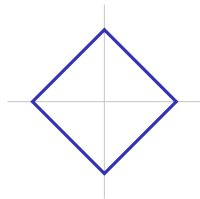
$$g(x) = (|x_1|^{1/2} + |x_2|^{1/2})^2.$$

Taking $x = e_1$, $y = e_2$ gives $g(x + y) = 4 > 2 = g(x) + g(y)$.

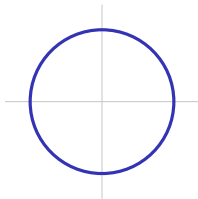
Key point: A useful nonnegative penalty need not be a norm. Test all three axioms.

The Unit Balls Look Different

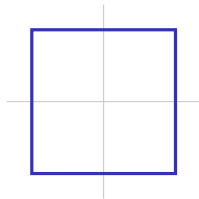
$\{x \in \mathbb{R}^2 : \|x\|_p = 1\}$ for $p = 1, 2, \infty$:



$$\|x\|_1 = 1$$



$$\|x\|_2 = 1$$



$$\|x\|_\infty = 1$$

The inclusions are visible: the 1-ball sits inside the 2-ball, which sits inside the ∞ -ball. That nesting is exactly the content of the inequalities on the next slides.

Key point: Different norms measure differently — but, as we are about to see, never by more than a factor depending only on n .

Hölder and Cauchy–Schwarz

Theorem (Hölder inequality [GVL (2.2.2)])

$$|x^T y| \leq \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

Cauchy–Schwarz [GVL (2.2.3)] — the case $p = q = 2$

$$|x^T y| \leq \|x\|_2 \|y\|_2.$$

Geometric reading

In $\mathbb{R}^3$, $|x^T y| = \|x\|_2 \|y\|_2 |\cos \theta|$ where θ is the angle between x and y . Cauchy–Schwarz is then just $|\cos \theta| \leq 1$. [GVL P2.2.10]

Key point: This is the inequality that lets an inner product be bounded by norms — the workhorse of every error estimate that follows.

Example 12: Bounds for a Dot Product

Take $x = (1, -2, 2)^T$ and $y = (2, 1, 2)^T$.

$$x^T y = 2 - 2 + 4 = 4, \quad \|x\|_2 = \|y\|_2 = 3.$$

Cauchy–Schwarz gives $|4| \leq 3 \cdot 3 = 9$. For $p = 1$ and $q = \infty$, Hölder gives

$$|x^T y| \leq \|x\|_1 \|y\|_\infty = 5 \cdot 2 = 10.$$

Attaining the bound. If $z = (1, -1, 1)^T$, then

$$x^T z = 5 = \|x\|_1 \|z\|_\infty.$$

Key point: A bound need not be tight for the given pair. Equality requires the directions or signs to align appropriately.

Cauchy–Schwarz: Proof by a Nonnegative Square

If $y = 0$, the result is immediate. Otherwise,

$$0 \leq \|x - ty\|_2^2 = \|x\|_2^2 - 2t(x^T y) + t^2\|y\|_2^2.$$

Choose $t_* = (x^T y)/\|y\|_2^2$, which minimizes the quadratic:

$$0 \leq \|x\|_2^2 - \frac{(x^T y)^2}{\|y\|_2^2}.$$

Multiply by $\|y\|_2^2$ and take square roots to obtain

$$|x^T y| \leq \|x\|_2 \|y\|_2.$$

Equality holds exactly when $x = t_* y$ (including the zero-vector cases).

Key point: The same minimizing scalar appeared in projection onto a line.

All Norms on $\mathbb{R}^n$ Are Equivalent

Theorem (Equivalence of norms [GVL (2.2.4)])

If $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are norms on $\mathbb{R}^n$, then there exist positive constants c_1, c_2 such that

$$c_1\|x\|_\alpha \leq \|x\|_\beta \leq c_2\|x\|_\alpha \quad \text{for all } x \in \mathbb{R}^n.$$

What this buys us

For a fixed finite dimension, all norms define the same convergence. A numerical tolerance must still account for the constants c_1, c_2 .

Consequence: convergence needs no norm [GVL §2.2.4]

$\{x^{(k)}\}$ converges to x if $\lim_{k \rightarrow \infty} \|x^{(k)} - x\| = 0$. By (2.2.4) this holds in **one** norm precisely when it holds in **all** of them — equivalently, componentwise.

Key point: The constants depend on n : in high dimension norms can still differ by a large factor, so “equivalent” is not “interchangeable in practice”.

The Explicit Constants

For $x \in \mathbb{R}^n$:

$$\|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2, \quad (2.2.5)$$

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty, \quad (2.2.6)$$

$$\|x\|_\infty \leq \|x\|_1 \leq n \|x\|_\infty. \quad (2.2.7)$$

	$\ x\ _1$	$\ x\ _2$	$\ x\ _\infty$
$x = e_1$ (one nonzero)	1	1	1
$x = (1, 1, \dots, 1)^T$ (spread out)	n	$\sqrt{n}$	1

The two rows are precisely the cases where the inequalities are attained.

Key point: The gap between the norms is largest for a vector whose mass is spread evenly, and vanishes for a vector concentrated in one component.

Deriving Norm Bounds and Their Equality Cases

Lower bound. Cross terms are nonnegative:

$$\|x\|_1^2 = \sum_i |x_i|^2 + 2 \sum_{i < j} |x_i| |x_j| \geq \|x\|_2^2.$$

Upper bound. Apply Cauchy–Schwarz to $(|x_i|)$ and $(1, \dots, 1)$:

$$\|x\|_1 \leq \sqrt{\sum_i |x_i|^2} \sqrt{\sum_i 1^2} = \sqrt{n} \|x\|_2.$$

For a nonzero vector, the lower bound is attained with one nonzero entry. The upper bound is attained when every $|x_i|$ is equal.

Check: For $(3, -4, 0)^T$, check $5 \leq 7 \leq 5\sqrt{3}$.

Example 13: Dimension Changes Numerical Tolerances

For $n = 10,000$, let $x_i = 10^{-3}$ for every entry.

$$\|x\|_{\infty} = 10^{-3}, \quad \|x\|_2 = 0.1, \quad \|x\|_1 = 10.$$

These values satisfy $\|x\|_1 = \sqrt{n}\|x\|_2 = n\|x\|_{\infty}$.

If these entries represent prediction errors,

$$\text{MAE} = \frac{\|x\|_1}{n} = 10^{-3}, \quad \text{RMSE} = \frac{\|x\|_2}{\sqrt{n}} = 10^{-3}.$$

Key point: State both the norm and any normalization. The same numerical tolerance can have different meanings at different dimensions.

Orthogonal Invariance of the 2-Norm

The 2-norm is preserved under orthogonal transformation

If $Q \in \mathbb{R}^{n \times n}$ is orthogonal and $x \in \mathbb{R}^n$, then $\|Qx\|_2 = \|x\|_2$.

Proof — three lines

$$\|Qx\|_2^2 = (Qx)^T(Qx) = x^T Q^T Qx = x^T x = \|x\|_2^2.$$

Why this is the single most useful property in the book

An exact orthogonal transformation preserves the 2-norm of an existing error vector. Computed transformations also introduce rounding errors, which need a separate stability analysis. This property supports stable Householder and Givens methods (§5.1).

Key point: For dimension at least two, other p -norms are not preserved by every orthogonal matrix.

Example 14: Rotation Preserves Only Certain Norms

Let $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ and $x = e_1$.

$$Q^T Q = I, \quad Qx = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Vector	1-norm	2-norm	Infinity-norm
x	1	1	1
Qx	$\sqrt{2}$	1	$1/\sqrt{2}$

For an existing error e , the *exact* transformation satisfies $\|Qe\|_2 = \|e\|_2$. A computed product has the form $\widehat{Qx} = Qx + \delta$ and can introduce a new rounding error δ .

Key point: Norm preservation is one ingredient in stability analysis, not a complete floating-point error proof.

Absolute and Relative Error

Suppose $\hat{x} \in \mathbb{R}^n$ approximates $x \in \mathbb{R}^n$.

Two measures

$$\epsilon_{\text{abs}} = \|\hat{x} - x\|$$

absolute error

$$\epsilon_{\text{rel}} = \frac{\|\hat{x} - x\|}{\|x\|}$$

relative error ($x \neq 0$)

Relative error accounts for the scale of the reference

Interpret absolute error using the units and application tolerance. For example, an error of 10^{-3} is excellent if $\|x\| \approx 10^6$ and catastrophic if $\|x\| \approx 10^{-6}$.

Relative Error and Significant Digits

For the book's example, compute the errors rather than rounding them to a power of ten:

$$x = \begin{bmatrix} 1.234 \\ 0.05674 \end{bmatrix}, \quad \hat{x} = \begin{bmatrix} 1.235 \\ 0.05128 \end{bmatrix}, \quad \hat{x} - x = \begin{bmatrix} 0.001 \\ -0.00546 \end{bmatrix}.$$

$$\begin{aligned} \|\hat{x} - x\|_{\infty} &= 0.00546, & \|x\|_{\infty} &= 1.234, \\ \epsilon_{\text{rel},\infty} &= 0.00546/1.234 \approx 0.004425. \end{aligned}$$

Component	Absolute error	Componentwise relative error
First	0.001	$0.001/1.234 \approx 0.0810\%$
Second	0.00546	$0.00546/0.05674 \approx 9.62\%$

Key point: The global error is about 0.4425%. It does not certify the same number of significant digits in every entry.

Example 15: A Small Global Error Can Hide a Bad Entry

Let $x = (10^6, 1)^T$ and $\hat{x} = (10^6, 0)^T$.

$$\|\hat{x} - x\|_{\infty} = 1, \quad \frac{\|\hat{x} - x\|_{\infty}}{\|x\|_{\infty}} = 10^{-6}.$$

The second entry has componentwise relative error $|0 - 1|/1 = 100\%$.

Near a zero reference. Relative error is undefined if $x = 0$. A common stopping rule combines absolute and relative scales:

$$\|\hat{x} - x\| \leq \text{atol} + \text{rtol} \|x\|.$$

Choose the tolerances using the units and accuracy required by the problem.

Key point: Global accuracy and componentwise accuracy answer different questions.

Convergence in Fixed and Changing Dimensions

Fixed space $\mathbb{R}^2$. Let $x^{(k)} = (1/k, (-1)^k/k)^T$. Then

$$\|x^{(k)}\|_1 = 2/k, \quad \|x^{(k)}\|_2 = \sqrt{2}/k, \quad \|x^{(k)}\|_\infty = 1/k.$$

All tend to zero even though the second coordinate changes sign.

A changing dimension. For $z^{(n)} = (1, \dots, 1)^T / \sqrt{n} \in \mathbb{R}^n$,

$$\|z^{(n)}\|_\infty = 1/\sqrt{n} \longrightarrow 0, \quad \|z^{(n)}\|_2 = 1.$$

This does not contradict finite-dimensional norm equivalence: there is no single fixed dimension.

Key point: Mesh refinement and increasing model size require attention to dimension-dependent constants and scaling.

Extension: A Norm Induced by a Linear Map

Define $\|x\|_C = \|Cx\|_2$ for a fixed matrix C . Homogeneity and the triangle inequality follow from the ordinary 2-norm. For positivity, we need

$$\|Cx\|_2 = 0 \implies x = 0,$$

which is equivalent to C having full column rank.

For $C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$, $\|x\|_C = \sqrt{x_1^2 + 4x_2^2}$ is a norm. For $D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\|De_2\|_2 = 0$ although $e_2 \neq 0$.

Key point: A rank-deficient map gives a seminorm: directions in its null space are invisible.

§2.2 — What to Carry Forward

Definitions

- Three axioms: positivity, triangle inequality, homogeneity
- p -norms; the trio 1, 2, ∞
- $\epsilon_{\text{rel}} = \|\hat{x} - x\|/\|x\|$

Inequalities

- Hölder; Cauchy–Schwarz
- Equivalence (2.2.4) with constants (2.2.5)–(2.2.7)

The two facts used most often

- 1 $\|Qx\|_2 = \|x\|_2$ for orthogonal Q — orthogonal transformations do not amplify error.
- 2 A normwise relative error is a global measure; inspect componentwise errors when small entries matter.

Next

§2.3 asks the same questions one level up: how do we measure a *matrix*?

Outline

- 1 §2.1 Basic Ideas from Linear Algebra
- 2 §2.2 Vector Norms
- 3 §2.3 Matrix Norms

Why Matrix Norms

The quality of a linear system solution may be poor if the matrix of coefficients is “nearly singular”. **To quantify the notion of near-singularity we need a measure of distance on the space of matrices.**

Definition (Matrix norm)

Since $\mathbb{R}^{m \times n}$ is isomorphic to $\mathbb{R}^{mn}$, the definition mirrors the vector case: $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a matrix norm if

$$\begin{aligned} f(A) &\geq 0, & A &\in \mathbb{R}^{m \times n}, \quad (f(A) = 0 \text{ iff } A = 0), \\ f(A + B) &\leq f(A) + f(B), & A, B &\in \mathbb{R}^{m \times n}, \\ f(\alpha A) &= |\alpha|f(A), & \alpha &\in \mathbb{R}, \quad A \in \mathbb{R}^{m \times n}. \end{aligned}$$

Key point: The axioms are identical to the vector case. Everything interesting comes from the *extra* properties some matrix norms have and others do not.

The Two Families

Frobenius norm [GVL (2.3.1)]

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$$

Treats A as a long vector of length mn .

Induced (operator) p -norms [GVL (2.3.2)]

$$\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p=1} \|Ax\|_p$$

Defined in terms of the *vector* p -norms of §2.2.

A family, not a norm

(2.3.2) defines a *family*: the 2-norm on $\mathbb{R}^{3 \times 2}$ is a different function from the 2-norm on $\mathbb{R}^{5 \times 6}$.

Two Consequences of the Induced Definition

1. The norm bounds the action of A

For all x ,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p.$$

$\|A\|_p$ is the largest factor by which A can stretch a vector.

2. Submultiplicativity [GVL (2.3.3)]

$$\|AB\|_p \leq \|A\|_p \|B\|_p, \quad A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times q}.$$

Strictly, this is a statement relating *three different norms*. Norms f_1, f_2, f_3 are *mutually consistent* if $f_1(AB) \leq f_2(A)f_3(B)$; in subscript-free notation,

$$\|AB\| \leq \|A\| \|B\|. \quad (2.3.4)$$

Key point: Submultiplicativity is what makes error bounds compose across a sequence of matrix operations.

Submultiplicativity: Every Step of the Proof

Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times q}$, and $x \in \mathbb{R}^q$. Apply the induced bound twice:

$$\|ABx\|_p \leq \|A\|_p \|Bx\|_p \leq \|A\|_p \|B\|_p \|x\|_p.$$

For $x \neq 0$, divide by $\|x\|_p$ and take the supremum:

$$\|AB\|_p = \sup_{x \neq 0} \frac{\|ABx\|_p}{\|x\|_p} \leq \|A\|_p \|B\|_p.$$

Interpretation. The worst possible stretch of a composition cannot exceed the product of the two worst stretches.

Check: Must one input vector attain both worst stretches simultaneously?

Not Every Matrix Norm Is Submultiplicative

A counterexample worth remembering

Define $\|A\|_{\Delta} = \max_{i,j} |a_{ij}|$. This satisfies all three norm axioms. Take

$$A = B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Then $AB = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, so

$$\|AB\|_{\Delta} = 2 > 1 = \|A\|_{\Delta} \|B\|_{\Delta}.$$

So $\|AB\|_{\Delta} > \|A\|_{\Delta} \|B\|_{\Delta}$: property (2.3.4) fails.

Key point: The entrywise maximum remains useful for componentwise analysis. Matrix products require a dimension factor or compatible input/output norms.

Subordinate Norms

More generally, mixing norms on the two spaces: for a vector norm $\|\cdot\|_\alpha$ on $\mathbb{R}^n$ and $\|\cdot\|_\beta$ on $\mathbb{R}^m$, define

$$\|A\|_{\alpha,\beta} = \sup_{x \neq 0} \frac{\|Ax\|_\beta}{\|x\|_\alpha}. \quad (2.3.5)$$

Then $\|Ax\|_\beta \leq \|A\|_{\alpha,\beta} \|x\|_\alpha$, and $\|\cdot\|_{\alpha,\beta}$ is said to be *subordinate* to the vector norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$.

The supremum is attained [GVL (2.3.6)]

Since $\{x \in \mathbb{R}^n : \|x\|_\alpha = 1\}$ is compact and $\|\cdot\|_\beta$ is continuous,

$$\|A\|_{\alpha,\beta} = \max_{\|x\|_\alpha=1} \|Ax\|_\beta = \|Ax_*\|_\beta$$

for some $x_* \in \mathbb{R}^n$ with unit α -norm.

Key point: “Sup” can be replaced by “max” — there is always a vector that achieves the norm.

Extension: A Mixed-Norm Example

With a 1-norm input and infinity-norm output,

$$\|A\|_{1,\infty} = \max_{i,j} |a_{ij}|.$$

Let $M = \max_{i,j} |a_{ij}|$. For every x ,

$$|(Ax)_i| \leq \sum_j |a_{ij}| |x_j| \leq M \|x\|_1.$$

If $|a_{i_* j_*}| = M$, choose $x = e_{j_*}$. Then $\|x\|_1 = 1$ and $\|Ax\|_\infty = M$.

Key point: The entrywise maximum has a clear operator meaning. Composition bounds require the output norm of one map to match the input norm of the next.

The Norms You Can Actually Compute

Maximum column sum [GVL (2.3.9)]

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$$

Maximum row sum [GVL (2.3.10)]

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$$

Example

$$A = \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix} \implies \|A\|_1 = \max\{3, 10\} = 10, \quad \|A\|_\infty = \max\{8, 5\} = 8.$$

Key point: Both cost $O(mn)$ for a dense m -by- n matrix. A general 2-norm calculation requires additional work.

Example 16: Column Sums and Row Sums

For $A = \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix}$, use absolute values before summing.

$$\text{Column sums: } |1| + |-2| = 3, \quad |-7| + |-3| = 10,$$

$$\text{Row sums: } |1| + |-7| = 8, \quad |-2| + |-3| = 5.$$

Hence $\|A\|_1 = 10$ and $\|A\|_\infty = 8$.

Vectors attaining the bounds.

$$Ae_2 = \begin{bmatrix} -7 \\ -3 \end{bmatrix}, \quad \|Ae_2\|_1 = 10; \quad A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \end{bmatrix}, \quad \left\| A \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\|_\infty = 8.$$

Both input vectors are unit vectors in their respective norms.

Why the Maximum Column Sum Is the Induced 1-Norm

Write $Ax = \sum_j x_j a_j$, where $a_j = A(:, j)$. Then

$$\begin{aligned}\|Ax\|_1 &\leq \sum_j |x_j| \|a_j\|_1 \\ &\leq \left(\max_j \|a_j\|_1 \right) \sum_j |x_j| = \left(\max_j \sum_i |a_{ij}| \right) \|x\|_1.\end{aligned}$$

This proves the upper bound on $\|A\|_1$. Choose a maximizing column j_* and set $x = e_{j_*}$:

$$\|x\|_1 = 1, \quad \|Ax\|_1 = \|a_{j_*}\|_1.$$

The upper bound is attained, so it equals the norm.

Key point: An induced-norm formula needs both an upper bound and a vector attaining it.

Why the Maximum Row Sum Is the Induced Infinity-Norm

For each row,

$$|(Ax)_i| \leq \sum_j |a_{ij}| |x_j| \leq \left(\sum_j |a_{ij}| \right) \|x\|_\infty.$$

Taking the largest row gives an upper bound. For a maximizing row i_* , choose $x_j = \text{sign}(a_{i_*j})$ (with $x_j = 1$ if the entry is zero). Then

$$\|x\|_\infty = 1, \quad (Ax)_{i_*} = \sum_j |a_{i_*j}|.$$

Thus the bound is attained: $\|A\|_\infty = \max_i \sum_j |a_{ij}|$.

Key point: Row signs remove cancellation at the selected output coordinate.

Relating the Matrix Norms

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{\min\{m, n\}} \|A\|_2, \quad (2.3.7)$$

$$\max_{i,j} |a_{ij}| \leq \|A\|_2 \leq \sqrt{mn} \max_{i,j} |a_{ij}|, \quad (2.3.8)$$

$$\frac{1}{\sqrt{n}} \|A\|_\infty \leq \|A\|_2 \leq \sqrt{m} \|A\|_\infty, \quad (2.3.11)$$

$$\frac{1}{\sqrt{m}} \|A\|_1 \leq \|A\|_2 \leq \sqrt{n} \|A\|_1. \quad (2.3.12)$$

Submatrices [GVL (2.3.13)] and convergence

$\|A(i_1:i_2, j_1:j_2)\|_p \leq \|A\|_p$ — a submatrix can never have a larger norm. And $\{A^{(k)}\} \rightarrow A$ means $\lim_{k \rightarrow \infty} \|A^{(k)} - A\| = 0$; as with vectors the choice of norm is immaterial.

Example 16: Frobenius Norm and Quick Bounds

For $A = \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix}$,

$$\|A\|_F = \sqrt{1^2 + (-7)^2 + (-2)^2 + (-3)^2} = \sqrt{63} \approx 7.9373.$$

The inexpensive estimates give

$$\|A\|_2 \leq \min\{\|A\|_F, \sqrt{\|A\|_1 \|A\|_\infty}\} = \min\{\sqrt{63}, \sqrt{80}\} = \sqrt{63}.$$

The column $Ae_2 = (-7, -3)^T$ gives a lower bound:

$$\|A\|_2 \geq \|Ae_2\|_2 = \sqrt{58} \approx 7.6158.$$

Key point: Before computing a spectral norm exactly, we already know it lies between 7.6158 and 7.9373.

Frobenius Norm: A Product Bound and a Distinction

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times q}$, Cauchy–Schwarz gives

$$\begin{aligned}\|AB\|_F^2 &= \sum_{i,j} \left| \sum_k a_{ik} b_{kj} \right|^2 \\ &\leq \sum_{i,j} \left(\sum_k |a_{ik}|^2 \right) \left(\sum_k |b_{kj}|^2 \right) = \|A\|_F^2 \|B\|_F^2.\end{aligned}$$

However, $\|I_n\|_F = \sqrt{n}$, whereas every same-norm induced operator norm satisfies $\|I_n\| = 1$.

Key point: The Frobenius norm is submultiplicative but, for $n > 1$, is not induced by using the same vector norm on the input and output.

Why Taking a Submatrix Cannot Increase an Induced p -Norm

Write a selected submatrix as $B = RAC$. Here C inserts zeros into the omitted input coordinates, and R selects output coordinates. For $1 \leq p \leq \infty$,

$$\|Cx\|_p = \|x\|_p, \quad \|Ry\|_p \leq \|y\|_p.$$

Therefore

$$\|Bx\|_p = \|RACx\|_p \leq \|ACx\|_p \leq \|A\|_p \|x\|_p.$$

Taking the supremum gives $\|B\|_p \leq \|A\|_p$.

Check: For the top-left entry of Example 16, check $|1| \leq \|A\|_2$.

Cost of Each Norm

Norm	Formula	Cost
$\ A\ _1$	max column sum [GVL (2.3.9)]	$O(mn)$, one pass
$\ A\ _\infty$	max row sum [GVL (2.3.10)]	$O(mn)$, one pass
$\ A\ _F$	root sum of squares [GVL (2.3.1)]	$O(mn)$, one pass
$\ A\ _2$	$\sqrt{\lambda_{\max}(A^T A)}$	SVD or iterative estimation

Key point: Three of the four are a single sweep over the entries. The general 2-norm uses a singular-value computation or an iterative estimate — which is why the bounds above matter.

The Matrix 2-Norm

Theorem (2.3.1)

For $A \in \mathbb{R}^{m \times n}$, a unit vector z exists such that $A^T A z = \|A\|_2^2 z$.

Derivation using the symmetric matrix $A^T A$.

$$\|A\|_2^2 = \max_{\|x\|_2=1} \|Ax\|_2^2 = \max_{\|x\|_2=1} x^T A^T A x = \lambda_{\max}(A^T A).$$

Thus $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}$.

Corollary 2.3.2: a cheap upper bound

$\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty}$, since

$$\mu^2 \|z\|_1 = \|A^T A z\|_1 \leq \|A^T\|_1 \|A\|_1 \|z\|_1 = \|A\|_\infty \|A\|_1 \|z\|_1.$$

Key point: Use $A^T A$ for hand derivations. A production calculation usually uses an SVD or matrix-vector iteration; explicitly forming $A^T A$ can worsen conditioning.

Example 16: Computing the Spectral Norm Exactly

For $A = \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix}$,

$$A^T A = \begin{bmatrix} 1 & -2 \\ -7 & -3 \end{bmatrix} \begin{bmatrix} 1 & -7 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ -1 & 58 \end{bmatrix}.$$

Its characteristic equation is

$$\begin{aligned} 0 &= (5 - \lambda)(58 - \lambda) - 1 \\ &= \lambda^2 - 63\lambda + 289. \end{aligned}$$

Thus

$$\lambda_{\pm} = \frac{63 \pm \sqrt{2813}}{2},$$

$$\|A\|_2 = \sqrt{\frac{63 + \sqrt{2813}}{2}} \approx 7.6170.$$

Key point: The eigenvalues used here belong to $A^T A$, not to A .

Example 16: A Vector Attaining the Spectral Norm

Let $\lambda_+ = (63 + \sqrt{2813})/2$. An eigenvector of $A^T A$ is

$$w = \begin{bmatrix} -1 \\ \lambda_+ - 5 \end{bmatrix}, \quad z = w/\|w\|_2 \approx \begin{bmatrix} -0.01886 \\ 0.99982 \end{bmatrix}.$$

Because $A^T A z = \lambda_+ z$ and $\|z\|_2 = 1$,

$$\|Az\|_2^2 = z^T A^T A z = \lambda_+, \quad \|Az\|_2 = \|A\|_2 \approx 7.6170.$$

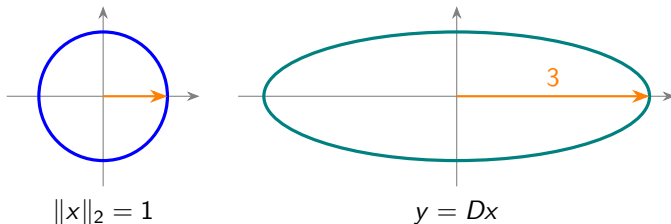
Norm	$\ A\ _1$	$\ A\ _\infty$	$\ A\ _F$	$\ A\ _2$
Value	10	8	$\sqrt{63}$	$\sqrt{(63 + \sqrt{2813})/2}$

Key point: The maximizing input is almost e_2 , explaining why the column-based lower bound was so accurate.

Example 17: The Unit Circle Becomes an Ellipse

For $D = \text{diag}(3, 1)$ and $x = (\cos \theta, \sin \theta)^T$,

$$Dx = (3 \cos \theta, \sin \theta)^T, \quad \|Dx\|_2^2 = 9 \cos^2 \theta + \sin^2 \theta.$$



Key point: $\|D\|_2 = 3$ measures the longest output radius. $\|D\|_F = \sqrt{10}$ combines both squared axis lengths.

Example 18: Eigenvalue Magnitudes Need Not Give the 2-Norm

Consider the nonsymmetric matrix

$$N = \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix}.$$

It has only the eigenvalue zero, since $\det(N - \lambda I) = \lambda^2$. Yet

$$Ne_2 = 5e_1, \quad N^T N = \begin{bmatrix} 0 & 0 \\ 0 & 25 \end{bmatrix}, \quad \|N\|_2 = 5.$$

For a real symmetric matrix, orthogonal diagonalization does imply $\|A\|_2 = \max_i |\lambda_i(A)|$.

Key point: Eigenvalues describe invariant directions. Singular values describe stretching of arbitrary directions.

Example 19: Estimating the Spectral Norm by Iteration

To estimate $\|A\|_2$, apply power iteration to $A^T A$ using products:

- 1 Choose x_0 with $\|x_0\|_2 = 1$.
- 2 Compute $y_k = Ax_k$ and $z_k = A^T y_k$.
- 3 If $z_k \neq 0$, set $x_{k+1} = z_k / \|z_k\|_2$.
- 4 Monitor $\mu_k = \|Ax_k\|_2$ and a suitable convergence criterion.

For a simple dominant eigenvalue and an initial vector with a nonzero component in its eigendirection, the iteration converges to that direction up to sign.

Key point: Each step needs Ax and $A^T y$, without storing $A^T A$. A single estimate is a lower bound, not automatically a certified upper bound.

Example 19: Four Hand-Checkable Iterates

Let $A = \text{diag}(3, 1)$ and $x_0 = (1, 1)^T / \sqrt{2}$. Since $A^T A = \text{diag}(9, 1)$,

$$x_k = \frac{(9^k, 1)^T}{\sqrt{9^{2k} + 1}}, \quad \mu_k = \sqrt{\frac{9 \cdot 9^{2k} + 1}{9^{2k} + 1}}.$$

k	$(x_k)_1$	$(x_k)_2$	μ_k
0	0.707107	0.707107	2.236068
1	0.993884	0.110432	2.983696
2	0.999924	0.012345	2.999797
3	0.999999	0.001372	2.999997

The exact answer is $\|A\|_2 = 3$. The unwanted coordinate ratio shrinks by $1/9$ per step.

Perturbations and the Inverse

Norms let us quantify how A^{-1} changes when A changes.

Lemma 2.3.3 (Neumann series)

If $F \in \mathbb{R}^{n \times n}$ and $\|F\|_p < 1$, then $I - F$ is nonsingular,

$$(I - F)^{-1} = \sum_{k=0}^{\infty} F^k, \quad \text{with} \quad \|(I - F)^{-1}\|_p \leq \frac{1}{1 - \|F\|_p}.$$

Proof idea

If $I - F$ were singular then $(I - F)x = 0$ for some $x \neq 0$, giving $\|x\|_p = \|Fx\|_p$ and hence $\|F\|_p \geq 1$ — a contradiction. The series then follows from $(\sum_{k=0}^N F^k)(I - F) = I - F^{N+1}$ and $F^k \rightarrow 0$.

Key point: An $O(\epsilon)$ perturbation of the identity induces an $O(\epsilon)$ perturbation of its inverse:

$$\|(I - F)^{-1} - I\|_p \leq \|F\|_p / (1 - \|F\|_p).$$

Neumann Series: The Finite Identity and the Error

Let $S_N = I + F + \cdots + F^N$ and $\|F\|_p = q < 1$. Multiplication telescopes:

$$S_N(I - F) = I - F^{N+1}.$$

Since $\|F^{N+1}\|_p \leq q^{N+1} \rightarrow 0$, the limit is $(I - F)^{-1}$. The remainder satisfies

$$\begin{aligned}\|(I - F)^{-1} - S_N\|_p &\leq \sum_{k=N+1}^{\infty} \|F^k\|_p \\ &\leq \sum_{k=N+1}^{\infty} q^k = \boxed{\frac{q^{N+1}}{1 - q}}.\end{aligned}$$

Key point: A convergence proof becomes an algorithmic stopping rule once the remainder is bounded.

Example 20: A Neumann Approximation with an Exact Error

Let $F = \text{diag}(1/2, 1/4)$, so $\|F\|_\infty = 1/2$ and

$$(I - F)^{-1} = \text{diag}(2, 4/3).$$

Use four terms, $S_3 = I + F + F^2 + F^3$:

$$S_3 = \text{diag}(1 + 1/2 + 1/4 + 1/8, 1 + 1/4 + 1/16 + 1/64) = \text{diag}(15/8, 85/64).$$

Thus

$$(I - F)^{-1} - S_3 = \text{diag}(1/8, 1/192), \quad \|(I - F)^{-1} - S_3\|_\infty = 1/8.$$

The bound is $(1/2)^4/(1 - 1/2) = 1/8$, exactly attained here.

Example 20: The Same Series as a Fixed-Point Solver

To solve $(I - F)x = b$, use $x_{k+1} = b + Fx_k$ with $x_0 = 0$. Then $x_k = (I + F + \dots + F^{k-1})b$. For $b = (1, 1)^T$ and the preceding F :

k	x_k	$\ x_* - x_k\ _\infty$
0	$(0, 0)^T$	2
1	$(1, 1)^T$	1
2	$(3/2, 5/4)^T$	1/2
3	$(7/4, 21/16)^T$	1/4
4	$(15/8, 85/64)^T$	1/8

Here $x_* = (2, 4/3)^T$ and $e_{k+1} = Fe_k$.

Key point: A matrix norm below one gives a uniform contraction bound for this iteration.

Extension: Sufficient Conditions Are Not Necessary Conditions

The norm test can fail while the series succeeds. For $F = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$, $\|F\|_2 = 2$, but $F^2 = 0$. Hence

$$(I - F)^{-1} = I + F = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

An inverse can exist while the series fails. For $F = 2I$, the matrix $I - F = -I$ is invertible, but $\sum_{k \geq 0} F^k$ diverges.

Key point: $\|F\| < 1$ is an easily checked sufficient condition. It is not equivalent to invertibility of $I - F$.

Perturbing a General Nonsingular Matrix

Theorem (2.3.4)

If A is nonsingular and $r = \|A^{-1}E\|_p < 1$, then $A + E$ is nonsingular and

$$\|(A + E)^{-1} - A^{-1}\|_p \leq \frac{\|E\|_p \|A^{-1}\|_p^2}{1 - r}.$$

Correct factorization and multiplication order. Set $G = A^{-1}E$:

$$\begin{aligned} A + E &= A(I + G), & (A + E)^{-1} &= (I + G)^{-1}A^{-1}, \\ (A + E)^{-1} - A^{-1} &= -(I + G)^{-1}GA^{-1}. \end{aligned}$$

Apply Lemma 2.3.3 with $F = -G$ and use $\|G\|_p \leq \|A^{-1}\|_p \|E\|_p$.

Key point: Keep the order of all factors. In general, $\|A^{-1}E\|_p$ and $\|EA^{-1}\|_p$ need not agree.

Example 21: Evaluating an Inverse-Perturbation Bound

Let $A = \text{diag}(2, 1)$ and $E = \text{diag}(0, -1/10)$.

$$\begin{aligned} A^{-1} &= \text{diag}(1/2, 1), & \|A^{-1}\|_{\infty} &= 1, \\ A^{-1}E &= \text{diag}(0, -1/10), & r &= 1/10 < 1. \end{aligned}$$

The perturbed inverse is $(A + E)^{-1} = \text{diag}(1/2, 10/9)$. Therefore

$$\|(A + E)^{-1} - A^{-1}\|_{\infty} = 1/9.$$

The theorem predicts

$$\frac{\|E\|_{\infty} \|A^{-1}\|_{\infty}^2}{1 - r} = \frac{(1/10) \cdot 1}{9/10} = 1/9.$$

Key point: Check $r < 1$ before using the conclusion. A perturbation bound can be sharp.

Preview: Small Residual and Large Solution Error

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 10^{-4} \end{bmatrix}, \quad x_* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b = Ax_* = \begin{bmatrix} 1 \\ 10^{-4} \end{bmatrix}, \quad \hat{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The residual $r = b - A\hat{x} = (0, 10^{-4})^T$ is small:

$$\frac{\|r\|_2}{\|b\|_2} \approx 10^{-4}.$$

But the relative solution error is

$$\frac{\|x_* - \hat{x}\|_2}{\|x_*\|_2} = \frac{1}{\sqrt{2}} \approx 70.7\%.$$

Since $x_* - \hat{x} = A^{-1}r$, weakly observed directions can amplify a residual.

Key point: Residual accuracy and solution accuracy are related through the inverse. This is the subject of §2.6.

Orthogonal Invariance — Again

If $A \in \mathbb{R}^{m \times n}$ and $Q \in \mathbb{R}^{m \times m}$, $Z \in \mathbb{R}^{n \times n}$ are orthogonal, then

$$\|QAZ\|_F = \|A\|_F, \quad (2.3.14)$$

$$\|QAZ\|_2 = \|A\|_2. \quad (2.3.15)$$

Proof for the Frobenius norm

Columnwise, by orthogonal invariance of the vector 2-norm,

$$\|QA\|_F^2 = \sum_j \|QA(:,j)\|_2^2 = \sum_j \|A(:,j)\|_2^2 = \|A\|_F^2; \text{ then } \|Q(AZ)\|_F^2 = \|AZ\|_F^2 = \|Z^T A^T\|_F^2 = \|A\|_F^2.$$

The most consequential fact in the chapter

Exact orthogonal transformations preserve both norms. This prevents amplification of existing errors at these steps; stability of a computed algorithm also requires control of newly introduced rounding errors.

Example 22: Matrix Norms Under a Rotation

Take $A = \text{diag}(3, 1)$ and $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

$$QA = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix}, \quad (QA)^T(QA) = A^T A = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}.$$

Consequently,

$$\|QA\|_2 = 3 = \|A\|_2, \quad \|QA\|_F = \sqrt{(9 + 1 + 9 + 1)/2} = \sqrt{10} = \|A\|_F.$$

However,

$$\|QA\|_1 = 3\sqrt{2}, \quad \|QA\|_\infty = 2\sqrt{2},$$

while $\|A\|_1 = \|A\|_\infty = 3$.

Key point: The 2-norm and Frobenius norm are orthogonally invariant; the 1-norm and infinity-norm generally are not.

Extension: The Norm of a Rank-One Matrix

Let $E = uv^T$ with nonzero vectors $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$. For every unit 2-norm vector x ,

$$\|Ex\|_2 = \|u\|_2 |v^T x| \leq \|u\|_2 \|v\|_2.$$

Equality is attained at $x = v/\|v\|_2$, so

$$\|uv^T\|_2 = \|u\|_2 \|v\|_2.$$

Also,

$$\|uv^T\|_F^2 = \sum_{i,j} u_i^2 v_j^2 = \left(\sum_i u_i^2 \right) \left(\sum_j v_j^2 \right).$$

Key point: For rank-one matrices, the Frobenius norm and the spectral norm are equal.

Extension: A Smallest-Norm Correction with a Constraint

We want $(A + E)s = y$ for a given $s \neq 0$. Put $r = y - As$; the constraint is $Es = r$. Every feasible E must satisfy

$$\|E\|_2 \geq \frac{\|Es\|_2}{\|s\|_2} = \frac{\|r\|_2}{\|s\|_2}.$$

Choose

$$E_* = \frac{rs^T}{s^T s}.$$

Then $E_*s = r$ and, using the rank-one formula,

$$\|E_*\|_2 = \frac{\|r\|_2\|s\|_2}{\|s\|_2^2} = \frac{\|r\|_2}{\|s\|_2}.$$

Key point: A universal lower bound plus an attaining construction proves optimality. The spectral-norm minimizer need not be unique.

Example 23: Computing a Minimum-Norm Correction

Let $A = I_2$, $s = (1, 2)^T$, and $y = (3, 1)^T$. Then $r = y - As = (2, -1)^T$ and $s^T s = 5$.

$$E_* = \frac{1}{5} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}.$$

Check the constraint:

$$E_* s = \frac{1}{5} \begin{bmatrix} 2+8 \\ -1-4 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} = r, \quad (I + E_*)s = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = y.$$

Since $\|r\|_2 = \|s\|_2 = \sqrt{5}$, $\|E_*\|_2 = \|E_*\|_F = 1$.

Key point: This is the smallest spectral norm achievable under the single constraint, not a guarantee about correcting every input.

§2.3 — What to Carry Forward

Definitions and structure

- Same three axioms as vector norms
- Frobenius (2.3.1); induced p -norms (2.3.2)
- $\|Ax\|_p \leq \|A\|_p \|x\|_p$ and $\|AB\| \leq \|A\| \|B\|$
- Not every norm is submultiplicative

Computation

- $\|A\|_1$: max column sum; $\|A\|_\infty$: max row sum
- $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}$ — SVD or iteration
- $\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty}$

Perturbation

- $\|F\| < 1 \Rightarrow I - F$ nonsingular,
 $\|(I - F)^{-1}\| \leq 1/(1 - \|F\|)$
- $\|(A + E)^{-1} - A^{-1}\| \leq \|E\| \|A^{-1}\|^2 / (1 - r)$

Invariance

$$\|QAZ\|_F = \|A\|_F, \quad \|QAZ\|_2 = \|A\|_2$$

The property that makes orthogonal methods the default throughout the book.

Where This Leads

§2.4 — The SVD

$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sigma_1$. The eigenvector z of Theorem 2.3.1 is the first right singular vector. The SVD delivers orthonormal bases for all four fundamental subspaces at once.

§2.5 — Subspace metrics

Uses the SVD and the orthogonality results of §2.1 to measure the distance *between subspaces*.

§2.6 — Sensitivity of $Ax = b$

Theorem 2.3.4 already exhibits the amplification factor $\|A^{-1}\|^2$. Section 2.6 turns it into the **condition number** $\kappa(A) = \|A\| \|A^{-1}\|$.

§2.7 — Finite precision

Combines the norms of §2.2–2.3 with a floating point model to give backward error analysis.

Key point: Every one of these depends on being able to *measure* — which is what §2.2 and §2.3 provide.

Practice: Three Calculations to Try Independently

- 1 For $C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, find a basis for its range and null space. Check rank-nullity.
- 2 For $A = I_2$, $u = v = (1, 1)^T$, and $b = (3, 0)^T$, solve $(A + uv^T)x = b$ using the rank-one update.
- 3 For $B = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$, compute $\|B\|_1$, $\|B\|_\infty$, $\|B\|_F$, and $\|B\|_2$.

Key point: For each answer, include a verification using the original definition or equation.

Practice Solution 1: Rank and Null Space

For $C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, the second row is twice the first. Thus $\text{rank} = 1$ and $\text{ran}(C) = \text{span}\{(1, 2)^T\}$. The single independent equation for $Cx = 0$ is

$$x_1 + 2x_2 + 3x_3 = 0.$$

Set $x_2 = s$ and $x_3 = t$:

$$x = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}.$$

The two displayed vectors are independent and are annihilated by C . Hence $\text{nullity} = 2$, and $1 + 2 = 3$, the number of columns.

Practice Solution 2: Sherman–Morrison

Here $y = A^{-1}b = (3, 0)^T$ and $z = A^{-1}u = (1, 1)^T$.

$$\alpha = 1 + v^T z = 3, \quad v^T y = 3.$$

Therefore

$$x = y - z \frac{v^T y}{\alpha} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Direct verification:

$$A + uv^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = b.$$

Check: Can you also find the inverse from the same update?

Practice Solution 3: Four Matrix Norms

For $B = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$, column sums are 1, 4 and row sums are 3, 2.

$$\|B\|_1 = 4, \quad \|B\|_\infty = 3, \quad \|B\|_F = \sqrt{1 + 4 + 0 + 4} = 3.$$

Moreover,

$$B^T B = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix}, \quad \det(B^T B - \lambda I) = \lambda^2 - 9\lambda + 4.$$

Thus

$$\|B\|_2 = \sqrt{\frac{9 + \sqrt{65}}{2}} \approx 2.9208.$$

The bound $\|B\|_2 \leq \min\{3, \sqrt{12}\} = 3$ is consistent.

Common Mistakes and a Final Check

- A basis for the original column space uses original pivot columns.
- A matrix can have repeated eigenvalues without enough eigenvectors.
- Matrix 1-norm means maximum absolute column sum, not sum of all entries.
- $\|A\|_2$ generally differs from both $\|A\|_F$ and $\max_i |\lambda_i(A)|$.
- $\|F\| < 1$ is sufficient for the Neumann series, not necessary for an inverse.
- Preserve multiplication order when differentiating or perturbing an inverse.
- Exact norm invariance does not eliminate rounding errors created by computation.

Check: Which of these mistakes can you disprove with a 2×2 example?

Reading and Selected Exercises

Primary reading

G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th edition, Johns Hopkins University Press, 2013, §§2.1–2.3, pp. 63–75.

Topic	Selected textbook exercises
Rank factorization and differentiation	P2.1.1–P2.1.3
Rayleigh quotient	P2.1.11
Norm inequalities and projection	P2.2.2, P2.2.4, P2.2.7–P2.2.8
Matrix-norm formulas	P2.3.1–P2.3.6
Rank-one and constrained corrections	P2.3.8–P2.3.9

The worked examples in this deck are teaching additions unless explicitly identified as the book's example.

Reference Card: Linear Algebra and Vector Norms

Linear algebra

$$\begin{aligned}\text{rank}(A) + \dim \text{null}(A) &= n, \\ \text{ran}(A)^\perp &= \text{null}(A^T), \\ (AB)^{-1} &= B^{-1}A^{-1}, \\ (A^{-1})' &= -A^{-1}A'A^{-1}.\end{aligned}$$

For $A = A^T$, $Q^T A Q = \text{diag}(\lambda_i)$ and

$$\lambda_{\min}(A) \leq \frac{x^T A x}{x^T x} \leq \lambda_{\max}(A).$$

Vector norms

$$\begin{aligned}\|x\|_1 &= \sum_i |x_i|, \\ \|x\|_2 &= \sqrt{x^T x}, \\ \|x\|_\infty &= \max_i |x_i|.\end{aligned}$$

$$|x^T y| \leq \|x\|_2 \|y\|_2,$$

$$\|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2,$$

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty.$$

For orthogonal Q , $\|Qx\|_2 = \|x\|_2$.

Reference Card: Matrix Norms and Perturbations

Matrix norms

$$\|A\|_1 = \max_j \sum_i |a_{ij}|,$$

$$\|A\|_\infty = \max_i \sum_j |a_{ij}|,$$

$$\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2},$$

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}.$$

$$\|Ax\|_p \leq \|A\|_p \|x\|_p, \quad \|AB\|_p \leq \|A\|_p \|B\|_p.$$

$$\|A\|_2 \leq \min\{\|A\|_F, \sqrt{\|A\|_1 \|A\|_\infty}\}.$$

Rank-one solve

$$y = A^{-1}b, \quad z = A^{-1}u,$$

$$\alpha = 1 + v^T z \neq 0, \quad x = y - z(v^T y)/\alpha.$$

Neumann series: $\|F\|_p < 1$

$$(I - F)^{-1} = \sum_{k=0}^{\infty} F^k,$$

$$\|(I - F)^{-1}\|_p \leq \frac{1}{1 - \|F\|_p}.$$

Inverse perturbation: $r = \|A^{-1}E\|_p < 1$

$$\|(A + E)^{-1} - A^{-1}\|_p \leq \frac{\|E\|_p \|A^{-1}\|_p^2}{1 - r}.$$

End of §2.1–2.3

Next: §2.4 The Singular Value Decomposition